

NGR Close Train Operating System

Role Profile System — Foundational Document v1.3

Updated April 2026 — incorporates field testing through firmware v1.3

Document History

v1.0 — Initial foundational document. Defined four stop profiles (DEFAULT, STATION, TRAILING, LEADER), role-based parameter system, and runtime adjustability via Dispatcher.

v1.1 — Safety corrections from field review. Added confirmed-stop debounce requirement for RELEASE, corrected dwell sequencing, gated leader departure on both dwell-expired AND RELEASE-received, added phase-lock prevention, and mandated decisive trailing stop.

v1.2 — Platform behavior introduced. Added MSG_TYPE_DEPARTING, trailing platform advance sequence (five sub-states), unified ramp rate, and differentiated leader/trailing ramp delays for positional control. Fixed deadlock caused by DEPARTING not being sent on STATION departures.

v1.3 — Full runtime tunability. Introduced rampUpRateMs as a named runtime-adjustable variable, advance PWM override for trailing platform approach, refined default timing values from field testing, and complete inline annotation of both loco and Dispatcher sketches.

Purpose

This document defines the behavioral model for locomotive stopping and departure in the NGR Close Train Operating (CTO) system. It establishes the four stop profiles, their intended behaviors, the tuning parameters that govern each, and the complete message protocol between locomotives and the Dispatcher. It is the reference document for all CTO profile-related firmware work.

Core Principle

A locomotive's stopping and departure behavior is determined entirely by its **role at the moment it stops**. Role is not fixed to a locomotive — it is assigned dynamically based on the locomotive's relationship to other traffic at each block boundary. The same locomotive may be TRAILING on one lap and LEADING on the next.

How Role Is Determined

When a locomotive detects a block boundary (two-magnet Hall sensor pair), the CTO logic queries the peer observation table before stopping:

Q1 — Am I closing on a leader?

Has another locomotive recently reported this same block and not yet moved beyond it? If yes, this locomotive is **TRAILING**. Q1 is evaluated first.

Q2 — Is someone closing on me?

Has another locomotive recently reported the block immediately behind mine? If yes, this locomotive is **LEADING**. Only evaluated if Q1 is false.

Neither

No close-running relationship detected. Normal station stop (STATION profile).

Role Unknown

No block has been detected yet, system just started, or trains have separated beyond the freshness window (45 seconds). DEFAULT profile applies — safe neutral operation, no waiting.

The Four Stop Profiles

DEFAULT

Condition: stopReason == STOP_NONE

Intent: Safe neutral operation. No encounter logic. No waiting for any other locomotive. Just keep the railroad moving. Applied at startup, when trains are separated, or whenever role cannot be determined. Does not include station hold behavior — that would cause unexpected holds at startup or during separation events.

Departure: Immediate on dwell expiry. No conditions.

STATION

Condition: stopReason == STOP_STATION_DWELL

Intent: Normal scheduled stop. Q1 and Q2 both returned false — no encounter relationship detected at this block. The locomotive waits for the configured dwell time. This profile implicitly accommodates a following train by giving it time to close the gap. DEPARTING is broadcast on departure so any trailing loco waiting at this block is unblocked.

Departure: Immediate on dwell expiry. DEPARTING broadcast.

TRAILING

Condition: stopReason == STOP_CTO_TRAILING

Intent: Stop immediately behind the leader. Once confirmed stop is achieved, send ENCOUNTER_START and RELEASE to the leader. Then enter the trailing platform advance sequence

— do NOT depart based on dwell alone.

Confirmed stop definition: PWM below 25 AND sustained for at least 250ms. RELEASE must not be sent before confirmed stop.

Trailing platform advance sequence (post-RELEASE):

- TR_WAIT_LEADER_DEPART — hold until MSG_TYPE_DEPARTING received from leader
- TR_FOLLOW_DELAY — wait followDelayMs after DEPARTING (default 3000ms)
- TR_ADVANCE_TO_PLATFORM — ramp to advancePwm (default cruisePwm/2), run for advanceMs (default 10000ms)
- TR_PLATFORM_STOP — ramp down, wait for confirmed stop
- TR_PLATFORM_DWELL — wait platformDwellMs (default 10000ms)
- Resume normal running — broadcast DEPARTING

Ramp delay: Zero — braking begins immediately on block detection. This gives the shortest possible stopping distance behind the leader.

LEADER

Condition: stopReason == STOP_CTO_LEADER_HOLD

Intent: Stop at the platform and hold. Wait for RELEASE from the trailing locomotive before departing. Dwell begins at confirmed physical stop and runs concurrently while waiting for RELEASE — it is not restarted or delayed when RELEASE arrives.

Ramp delay: 6000ms (default) — the leader coasts 6 seconds past the block detection point before braking. This positions the leader well into the platform area, creating separation from the trailing loco's stop point.

Departure gate — both conditions must be true:

- Dwell time has expired
- Valid RELEASE received from the correct trailing locomotive with matching encounter ID

■ Neither condition alone is sufficient. RELEASE before dwell expiry does not trigger departure. Dwell expiry without RELEASE does not trigger departure. Both must be satisfied.

On departure: Send MSG_TYPE_DEPARTING to the trailing locomotive. This is the gate signal that begins the trailing loco's platform advance sequence.

MSG_TYPE_DEPARTING — Critical Protocol Addition

Added in v1.2. Sent by any loco at the exact moment rampTarget is set positive after a stop. This message is the gate that allows the trailing loco to begin its platform advance. Without it, the trailing loco would wait indefinitely.

Key rule: DEPARTING must be broadcast on **every** departure — not just LEADER departures. If a trailing loco arrives at a block after the leader has already committed to a STATION stop (no encounter detected at that moment), the leader will send no RELEASE-gated DEPARTING. The STATION departure broadcast unblocks the trailing loco. Without this, deadlock occurs.

The DEPARTING packet carries the sender's locold, blockId, and encounterId. When sent from a STATION or DEFAULT departure, encounterId is 0 and locold target is 0 (broadcast). The trailing loco accepts any DEPARTING from its known leader at the matching block, regardless of encounter ID.

Confirmed Stop — Critical Definition

A locomotive is considered fully stopped when:

- PWM (rampCurrent) is at or below 25, AND
- That condition has been sustained for at least 250ms continuously

■ Sending RELEASE before physical stop allows the leader to depart while the trailing locomotive is still rolling — a rear-end collision risk. The debounce requirement is non-negotiable. If PWM rises above 25 during the debounce window, the timer resets.

Dwell Sequencing

- Dwell starts at confirmed physical stop — not at block detection, not at RELEASE receipt
- Dwell runs concurrently — LEADER dwell runs while waiting for RELEASE, not after
- Dwell is a minimum, not a trigger — expiry opens the gate but does not force departure
- TRAILING dwell is minimal (500ms default) — departure is gated on DEPARTING, not dwell

Ramp System

All PWM changes — acceleration and deceleration — are managed exclusively by serviceRamp(). No other function calls ledcWrite() directly. This ensures smooth, predictable motion and prevents abrupt changes.

Deceleration (rampRateMs)

Milliseconds between each PWM decrement during braking. Applied from the active profile (role-specific). Higher = slower, gentler braking over longer distance. Lower = harder, shorter stop. In v1.3 this is a shared value across all profiles — one physical track, one braking characteristic. Adjustable via **rr####**.

Acceleration (rampUpRateMs)

Milliseconds between each PWM increment during acceleration. Applied identically to all departures regardless of role. Higher = slower, gentler ramp-up. Default 120ms/step. Adjustable via **ru####**. This is

the primary control for how smoothly locos pull away from stops.

Ramp Delay (rampDelayMs)

Milliseconds the loco coasts at cruise speed after block detection before braking begins. Controls physical stop position relative to the detection point. Leader: 6000ms default (stops well into platform). Trailing: 0ms (stops immediately). Adjustable per role via Dispatcher.

Trailing Platform Advance Parameters

These parameters control the trailing loco's journey to the platform after RELEASE. All are runtime-adjustable from the Dispatcher without reflashing.

Parameter	Default	Command	Description
followDelayMs	3000ms	bfd3000	Wait after DEPARTING before advancing
advanceMs	10000ms	bad10000	Time spent moving toward platform
advancePwm	cruise/2	bap60	PWM during advance (override 0=auto)
platformDwellMs	10000ms	bpd10000	Time stopped at platform

advanceMs must compensate for the leader's rampDelayMs at half advance speed. At cruisePwm/2 with ramp time, 10000ms approximates the distance covered by the leader during its 6000ms coast. Tune from the Dispatcher — no reflash needed.

Complete Tuning Parameter Reference

Parameter	Scope	Dispatcher Command	Default
Cruise PWM	Per loco	op### / tp### / bp###	Config file
Ramp delay	Per role	[o t b][role]rd####	See profiles
Ramp rate (decel)	All profiles	rr###	80ms/step
Ramp up rate	All depart.	ru###	120ms/step
Dwell	Per role	[o t b][role]dw####	See profiles
Follow delay	Trailing	[o t b]fd#####	3000ms
Advance duration	Trailing	[o t b]ad#####	10000ms
Advance PWM	Trailing	[o t b]ap###	cruisePwm/2
Platform dwell	Trailing	[o t b]pd#####	10000ms

Role abbreviations: df=DEFAULT st=STATION tr=TRAILING ld=LEADER. Target abbreviations: o=Otto t=Toby b=both. All changes take effect immediately and reset to compiled defaults on power cycle.

Compiled Default Profile Values

Profile	Ramp Delay	Ramp Rate	Dwell	Notes
DEFAULT	200ms	80ms/step	2000ms	No encounter logic
STATION	200ms	80ms/step	5000ms	Normal stop, DEPARTING on exit
TRAILING	0ms	53ms/step	500ms	Immediate brake; departs via sub-states
LEADER	6000ms	80ms/step	4000ms	Coasts to platform; holds for RELEASE

rampUpRateMs = 120ms/step (all profiles). trailingAdvancePwm = cruisePwm/2 (auto).

ESP-NOW Message Protocol

All communication uses a fixed 28-byte packed struct broadcast to MAC FF:FF:FF:FF:FF:FF. Every node receives every packet. Nodes filter by magic (0x4E), version (6), and senderId (own packets are discarded). Packet VERSION must match across all three nodes — mismatches are silently dropped.

Message	Sender	Key recipient action
MSG_TYPE_BLOCK	Any loco	All nodes update peer table with block/timestamp
MSG_TYPE_ENCOUNTER_START	Trailing	Leader notes trailing has arrived at block
MSG_TYPE_RELEASE	Trailing	Leader sets leaderReleaseReceived=true
MSG_TYPE_DEPARTING	Any loco	Trailing sets departingReceived=true; begins follow delay
MSG_TYPE_ESTOP	Dispatcher	All locos: running=false, rampTarget=0, state=STOP_ESTOP
MSG_TYPE_MANUAL	Dispatcher	Locos act on embedded CmdType and parameters

Typical Two-Train Encounter Sequence

The behavioral result of a correctly functioning CTO encounter at a block:

Step	Event
1	Leader arrives at block, coasts 6s, stops at platform (LEADER profile)
2	Trailing arrives, brakes immediately, stops behind leader (TRAILING profile)
3	Trailing: confirmed stop → sends ENCOUNTER_START + RELEASE to leader
4	Leader: dwell expired + RELEASE received → sends DEPARTING → departs
5	Trailing: DEPARTING received → follow delay 3s
6	Trailing: advances to platform at cruisePwm/2 for 10s

7	Trailing: platform stop (confirmed) → platform dwell 10s
8	Trailing: departs, broadcasts DEPARTING → resumes normal running

E-STOP

E-STOP is not a profile. It overrides all profile and encounter logic immediately. A locomotive receiving E-STOP sets `running=false`, `rampTarget=0`, `stopReason=STOP_ESTOP` and will not respond to `CMD_START` until power-cycled or manually reset. E-STOP is broadcast to all nodes simultaneously.

What Remains Locomotive-Specific

Cruise PWM — each locomotive may require a different cruise speed reflecting its motor characteristics, wheel condition, and weight. This is the only parameter that remains per-locomotive. All other tuning parameters are role-based and apply identically to both locos.

Deferred — Future Versions

- Role lock at block boundary — prevent role flipping near detection edge cases
- Randomized dwell — small variation per stop to prevent long-term synchronization
- Persistent parameter storage across power cycles (SPIFFS/NVS)
- Self-calibration of stopping distance based on measured overshoot
- Sensor-based platform precision — timed approach currently used
- Support for more than two locomotives
- Full Work Zone integration — currently stubbed in Dispatcher

Relationship to CTO Architecture

The `stopReason` enum is the profile selection hook — set at block detection before any profile values are accessed. Profiles are selected in `startDwellAtBlock()` and applied throughout `serviceDwelling()` and `serviceRamp()`.

`serviceRamp()` is the sole owner of `ledcWrite()`. All other code changes `rampTarget` only. `serviceRamp()` moves `rampCurrent` toward `rampTarget` one step per interval, using `rampRateMs` for deceleration and `rampUpRateMs` for acceleration.

The trailing sub-state machine runs inside `serviceDwelling()` after confirmed stop and `RELEASE` have been sent. It is a separate sequential state machine that runs to completion before normal running resumes. The main `detectState` remains `DWELLING` throughout the entire trailing sequence.

NGR Lowline — CTO Role Profile System — Foundational Document v1.3 — April 2026 — Field-tested through firmware v1.3